The D. G. Ruparel College
OF ARTS, SCIENCE AND COMMERCE
Department of Computer Science ▪ Information Technology

Name : Sahil Tukaram Sutar
Class : TYBSc.IT Batch : D Expt. No. : 19
Roll No. : IT-6236 Date : Signature :
TITLE :

Aim:- Design and implement an embedded access-control prototype that grants entry only when either a whitelisted RFID tag is detected or a legitimate fingerprint is verified, provides real-time visual feedback on an LCD, indicates status via an LED, and pulses a relay (active-LOW) to unlock for a fixed interval. The system must log all events (time, method, result) for auditability.

* Hardware Requirements
1. Raspberry Pi Model A/B/B+
2. Jumper wires (Female to Male)
3. Relay Module
4. RFID RC522
5. R307/R305 fingerprint sensor
6. 16X2 LCD

* Software Requirements
1. Raspbian Stretch OS

1. Connect with Raspberry Pi's GPIO Pins.

- Relay Module

Device / Pin	Raspberry Pi 4	Physical Pin
Relay VCC	5V	2
Relay GND	GND	9
Relay IN	GPIO 23	16

* LCD 16x2 Connections

Device / Pin	Raspberry Pi	Physical Pin
LCD GND	GND	6
LCD VCC	5V	4
LCD SDA	GPIO2 / SDA	3
LCD SCL	GPIO3 / SCL	5

* R307 / R305 Fingerprint

R307S Wire	R307S Function	Raspberry Pin
Red	VCC / 5V	Pin 2
Black	GND	Pin 6
Yellow	TXD	Pin 10
Green	RXD	Pin 8

*RC522 RFID → Raspberry Pi 4

RC522 Pin	Raspberry Pi Physical Pin	GPIO
3.3V	Pin 1	3.3V
RST	Pin 22	GPIO25
GND	Pin 6	GND
MISO	Pin 21	GPIO9
MOSI	Pin 19	GPIO10
SCK	Pin 23	GPIO 11
SDA/SS	Pin 24	GPIO 8

*Software Installation

```
sudo apt update
sudo apt install -y python3-pip python3-serial
    python3-rpi.gpio i2c-tools pip3 install
    RPLCD adafruit-circuitpython-fingerprint
      mfrc522

sudo raspi-config
Interface Options → I2C

→Enable sudo i2c-dete i2cdetect

-y 1
```

I2C LCD Address

The LCD shown in the supplied image uses an I2C backpack. The common address is 0x27. If i2cdetect shows 0x3F, change the LCD address in the program from 0x27 to 0x3F.

```
#Update system repositories
sudo apt update

#install required system tools and libraries.
sudo apt install -y python3-pip python3-serial
  python3-rpi.gpio i2c-tools.

#install Python modules using
---break-system-packages (required for Raspberry
 Pi OS Bookworm)
PIP3 install RPLCD adafruit-circuitpython-
   fingerprint mfrc522
---break-system-package.

sudo i2cdetect -y 1
```

* Code:-

Program 1:- RFID (rfid_test.py)

```
from mfrc522 import SimpleMFRC522
import RPi.GPIO as GPIO

reader = SimpleMFRC522()

try:
    print("Place RFID card...")
    while True:
        uid, text = reader.read()
        print("RFID UID:", uid)
        print("Data:", text)
        break
finally:
    GPIO.cleanup()
```

Program 2: Fingerprint (enroll.py)

```
import serial
import adafruit_fingerprint

#UART Configuration on Raspberry Pi GPIO 14/15
uart = serial.Serial("/dev/serial0", baudrate=57600,
                                    timeout=1)
```

```
finger = adafruit_fingerprint.Adafruit_Fingerprint
                                (uart)

location = int(input("Enter fingerprint ID (1-127): "))

print("Place finger...")
while finger.get_image() != adafruit_fingerprint.
                                OK:
    pass

if finger.image_2_tz(1) != adafruit_fingerprint.
                                OK:
    print("Image conversion failed")
    exit()

print("Remove finger")
input("Press Enter after removing finger...")

print("Place same finger again...")
while finger.get_image() != adafruit_fingerprint.
                                OK:
    pass

if finger.image_2_tz(2) != adafruit_fingerprint.OK:
    print("Image conversion failed")
    exit()
```

```
if finger.create_model() != adafruit_fingerprint.OK:
    print("Fingerprint mismatch")
    exist()

if finger.store_model(location) == adafruit_fingerprint
                                        .OK:
    print("Fingerprint enrolled successfully")
else:
    print("Enrollment failed")
```

Program 3:- main program (access_control.py)

```
import RPi.GPIO as GPIO
import time
import csv
import os
import serial
from datetime import datetime
from RPLCD.i2c import CharLCD
from mfrc522
import SimpleMFRC522
import adafruit_fingerprint

#Pin Definitions (BCM Numbering)
RELAY = 23
```

```
LED = 21

GPIO.setmode(GPIO.BCM)
GPIO.setup(RELAY, GPIO.OUT)
GPIO.setup(LED, GPIO.OUT)

#Active-LOW Relay: default HIGH (OFF)
GPIO.output(RELAY, GPIO.HIGH)
GPIO.output(LED, GPIO.LOW)

#Initialize I2C LCD (change 0x27 to 0x3F if
 i2cdetect reported 0x3F)
lcd = CharLCD(
   i2c_expander='PCF8574',
   address=0x27,
   port=1,
   cols=16,
   rows=2,
   charmap='A00',
   auto_linebreaks=True
)

reader = SimpleMFRC522()

# Replace with the decimal UID
obtained from rfid_test.py
```

```
WHITELIST = [
    12345678,
    98765432
]

#Initialize Fingerprint Sensor
uart = serial.Serial("/dev/serial0", baudrate =
                     57600, timeout=1)
finger = adafruit_fingerprint.Adafruit_finger-
                     print(uart)

LOG_FILE = "access-log.csv"

def log_event(method, result):
    new_file = not os.path.exist(LOG_FILE) with
        open(LOG_FILE, "a", newline="") as file
    writer = csv.write(file)
    if new_file:
        write.writerow(["Time", "Method", "Result"])
    writer.writerow([datetime.now().strftime
        ("%Y-%m-%d %H:%M:%S"),
            method, result])
            re
def show_message(line1, line2=""):
    lcd.clear()
    lcd.write_string(line1[:16])
```

```
    if line2:
        lcd.cursor_pos = (1,0)
        lcd.write_string(line2[:16])

def unlock(method):
    show_message("ACCESS GRANTED", method)
    GPIO.output(LED, GPIO.LOW) HIGH)
    GPIO.output(RELAY, GPIO.LOW)
#Active-Low trigger
    log_event(method, "GRANTED")
    time.sleep(5)
    GPIO.output(RELAY, GPIO.HIGH)
# Turn off relay
    GPIO.output(LED, GPIO.LOW)

def deny(method):
    show_message("ACCESS DENIED", method)
    GPIO.output(LED, GPIO.LOW)
    log_event(method, "DENIED")
    time.sleep(2)

def check_fingerprint():
 # Poll sensor for up to 3 seconds
 start = time.time()
 while time.time() - start < 3:
     result = finger.get_image()
```

```
        if result == adafruit_fingerprint.OK:
            break
        elif result == adafruit_fingerprint.NOFINGER:
            time.sleep(0.1)
            continue
        else:
            return False
    else:
        return False

    show_message("Processing...")
    if finger.image_2_tz(1) != adafruit_fingerprint.OK:
        return False
    if finger.finger_search() != adafruit_fingerprint
                                    .OK:
        return False
    return True

try:
    show_message("ACCESS CONTROL", "SYSTEM READY
                                    ")
    time.sleep(2)

    while True:
        show_message("Scan RFID", "or Fingerprint")
```

```
        #Non-blocking RFID check
        card_id, text = reader.read_no_block()
        if card_id:
            print("RFID UID:", card_id)
            if card_id in WHITELIST:
                unlock("RFID")
            else:
                deny("RFID")
            time.sleep(1)
            continue

        #Check fingerprint sensor
        if check_fingerprint():
            print("Fingerprint matched")
            unlock("FINGERPRINT")
            time.sleep(1)

    except KeyboardInterrupt:
        print("\nProgram stopped")
    finally:
        GPIO.output(RELAY, GPIO.HIGH)
        GPIO.output(LED, GPIO.LOW)
        GPIO.cleanup()
        lcd.clear()
```

*Run the Program

```
python3 rfid_test.py
python3 enroll.py
python3 access_control.py
```